Henry Shields

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Education

Northeastern University, Boston, MA, GPA 3.97 **Dec 2026**

Bachelor of Science in Computer Engineering and Computer Science with a Minor in Mathematics

***Coursework**:* Object-Oriented Design, Algorithms and Data Structures, Computer Systems, Circuits and Signals, Databases, Embedded and Digital Design, Software Engineering, Electronics, Mobile Robotics, Quantum Engineering, and Networks

***Activities**:* NUWireless, SEDS, Roxbury Robotics**,** IEEE, Mathematics at the Alfréd Rényi Institute in Budapest

Skills

***Languages & Frameworks**:* C/C++, C#, Java, Python, ROS2, MATLAB, MoveIt, HTML/CSS, SQL, Assembly, Verilog

***Development Tools**:* Git/GitHub, IsaacSim, Unity, PSoC Creator, AutoCAD, Code Composer Studio, KiCad, Linux, LTSpice, MCUXpresso, SOLIDWORKS, VS Code, Windows, Xilinx Vivado

***Hardware & Electronics**:* Robotics Systems Integration (Servo Control, Motion Planning, Trajectory Creation), FPGA Development, Signal Processing (I2S, PWM, UART, BLE), Arduino, Circuit Design, Lab Equipment (DMM, Oscilloscope, Waveform Generator), PCB Programming (BSL, JTAG, USB-C), Soldering

Work Experience

Chewy – Boston, MA **Jun – Dec 2025**

Robotics Software Engineer I Co-op

- Designed and implemented leader-follower teleoperation system with direct firmware/software control, developing Python-based automated calibration scripts and PID-optimized arm stiffness control for real-time bimanual synchronization using custom 6-DOF scaled arm prototypes.
- Engineered complete hardware integration and system optimization including Dynamixel servo integration, software-based controller tuning to eliminate jitter, real-time data recording systems, intuitive teleoperation GUI, and comprehensive launch automation for seamless multi-robot coordination that reduces setup time by 85%.
- Developed keyboard teleoperation system using ROS2, MoveIt, and MoveIt Servo for dual-arm UR16e control with Isaac Sim simulation, implementing normalized gripper control interface for modular end-effector compatibility.
- Modified UR Driver codebase to support N concurrent robot instances, enabling simultaneous control of dual-arm systems and expanding system scalability.

Viomerse Inc. – Rochester, NY **May – Dec 2024**

Firmware and Software Engineer for the VioVision XR Display

- Designed and executed novel firmware features in C++ including reflashing, menuing, and calibration modes while optimizing Flash memory for an increased 70% space through refactoring and abstraction.
- Implemented a new USB-C board programming system in BSL mode, enabling non-invasive headset programming and streamlining the flashing process by 2000% as verified by the CTO.
- Self-taught Unity and C# to develop comprehensive XR calibration wizard and in-headset HUD menu system, implementing stereoscopic calibration processes, real-time settings control, and Zoom API integration.
- Engineered complete PDM Microphone to I2S to USB-A Audio system using Cypress PSoC 6 BLE.
- Developed local network infrastructure using NXP WiFi 6 MCUs for LAN connectivity and wireless streaming.
- Authored research paper on near-eye vs standard display effective resolutions and comprehensive technical documentation following ISO 13485/IEC 62304 standards.

Projects

ProPlay Self-Training Baseball Bat **Dec 2025 – Present**

- Enhanced multi-thousand-line C++ firmware for ESP32-based IoT sports device, implementing gyroscope-accelerometer fusion and gesture recognition that vastly improved user experience and consistency.
- Optimized power management with motion interrupt-based wake-up and resolved critical bugs in state machine logic, audio playback, and battery systems through direct client collaboration.

ECG Signal Acquisition and Processing System **Mar – Apr 2024**

- Created an ECG data acquisition system with active filters (LT1490) and instrumentation amplifier (AD627).
- Implemented multi-stage digital filtering algorithms in MATLAB for enhanced ECG signal quality.

Processor Design and Implementation Project **Feb – Apr 2024**

- Utilized SystemVerilog HDL on Xilinx Vivado to create a fully functional 16-bit RISC-V processor.
- Constructed components including 16-bit ALU, register file, data/instruction memory, and control logic.